**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

# The VFD Keeps Tripping: Reading the Fault Code

Field notes from Jonathan Gilmour, independent controls engineer, Bay Area and Northern California.

A variable frequency drive (VFD) that trips is doing its job. It saw something outside a limit and shut the motor down before it damaged the motor, the drive, or the machine. So the first mistake I see at 2am is treating the trip as the problem. The trip is the drive telling you something. The fault code is the drive telling you *what* it saw. Your job is to read what it saw, figure out why it saw it, and fix the cause. Not to reach into the parameters and widen the limit so the drive stops complaining. That last move gets motors burned and, on a bad day, gets a coupling let go with someone standing next to it.

This guide is organized by fault class, because that is how the drive thinks and that is how you should work it. Overcurrent means one set of things. Overvoltage means a different set. The code narrows the search before you ever pick up a meter. What follows works across sectors, whether the drive is on a conveyor, a pump, a mixer, a fan, or a blower feeding a digester. I will flag where a corrosive or gas plant changes the answer.

## Before you touch anything: read the log

Every modern drive keeps a fault log with a timestamp and, on most platforms, a snapshot of the conditions at the moment of the trip. That snapshot is the single most useful thing on the machine, and most people walk right past it. Before you change one parameter, pull the log and answer these:

- What is the exact fault code, and what does the manual for *this* drive family call it? Codes are not universal between manufacturers. F005 on one brand is not F005 on another.
- What was the drive doing at the trip? Accelerating, decelerating, or running at steady speed? Overcurrent on accel and overcurrent at constant speed point at different causes.
- Was it loaded? Look at the output current and torque in the snapshot if the drive stores them.
- What time did it trip, and is there a pattern? A drive that trips every afternoon in July is telling you about ambient heat. A drive that trips when the line startup sequence energizes three big loads at once is telling you about line sag. A drive that trips right when an overhead crane runs nearby is telling you about noise or line disturbance.

- How many times, and does it auto-restart? Repeated trips with auto-restart cycling can mask the original event. Find the first one.

Write down the code, the timestamp, and the conditions before you do anything else. If you change a parameter first, you have destroyed the evidence and you are now guessing.

## Safety, before the covers come off

Most VFD diagnosis is done with the drive powered and running, watching the display and the load. That is fine at the keypad. It stops being fine the moment you open the enclosure or put probes on line or motor terminals.

WARNING — stored energy. A VFD has a DC bus with large capacitors that hold a lethal charge after you remove power. Do not trust that "off" means "safe." After opening the disconnect, wait the full discharge time the manufacturer prints on the drive (commonly several minutes, often 5, sometimes more on larger frames), then *verify* the DC bus is below a safe level with a meter across the two bus terminals (commonly labeled DC+ / +DC and DC- / -DC; on some drives the brake terminals sit on the same DC+ rail, but read the bus across DC+ to DC-, and confirm the labels in this drive's manual) before your hands go near anything. Verify, do not assume.

WARNING — energized work. Any measurement on live line or motor terminals is energized electrical work. That means the right personal protective equipment (PPE) for the available incident energy, a rated meter, and ideally a second person. If your site has an arc-flash label on the panel, follow it. If it does not, treat the panel as if it can hurt you, because it can.

Lockout/tagout (LOTO). Before any hands-on work on the motor, the coupling, the belt, or anything the motor drives — full LOTO to a zero-energy state. A drive can be commanded to start remotely, over a network, by a sequence you did not know was armed. Rotating equipment does not care that you were only going to be a second.

Gas and confined-space plants. If this drive runs a blower, pump, or mixer tied to a digester, a wet well, a sump, a gas holder, or a pump vault, that space is a permit-required confined space. Hydrogen sulfide (H2S) kills fast and deadens your sense of smell around 100 parts per million (ppm), which is also roughly its immediately-dangerous-to-life-or-health (IDLH) level, so smell tells you nothing about safety. Methane (CH4) is explosive from about 5% by volume (lower explosive limit, LEL) to about 15% (upper explosive limit, UEL). Test the atmosphere before entry and wear a bump-tested personal 4-gas monitor (H2S, LEL, oxygen, carbon monoxide). Biogas areas are typically classified Class I Division 1 or 2 (or Zone 0/1/2) under the National Electrical Code (NEC) Articles 500/505; the drive enclosure and your tools must suit the classification. If the drive itself sits in a classified area, do not open it live, period.

## The fault classes, and what each really means

### Overcurrent (instantaneous overcurrent, IOC)

**What it means:** the drive's output current exceeded a hard, fast limit — usually well above the motor's full-load amps (FLA) — in a very short window. This is a fast trip, not a slow thermal one. The drive shuts down in milliseconds to protect its output transistors.

**What causes it, in check order:**

- A short or ground in the motor cable or motor windings. This is the safest-first assumption because it is the most dangerous to ignore. A phase-to-phase short or a phase-to-ground fault will trip IOC the instant the drive fires. If the drive trips IOC immediately on start, before the motor even turns, suspect the cable or motor first.
- Mechanical bind or jam. The load is locked — a jammed conveyor, a seized bearing, a pump against a closed valve, frozen product. The motor tries to break it loose, current spikes, IOC trips. This one usually trips right at the start of accel.
- Acceleration too fast. The ramp asks the motor to change speed faster than the load's inertia allows, current climbs to make the torque, and it hits the limit. This is a real cause and it is *also* the one people abuse — they lengthen the ramp until the trip stops even though the real problem was a bind. Lengthen the ramp only after you have ruled out a mechanical cause.
- Output devices failing in the drive itself, on older or heat-stressed units.

**How to check:** With the drive isolated and LOTO applied and the DC bus verified dead, disconnect the motor leads at the drive and check the motor and cable insulation with a megohmmeter (megger) — see Ground Fault below for how. Turn the shaft by hand (once safe) to feel for a bind. Compare the drive's set accel time against what the machine actually needs. Do not raise the current limit to make IOC go away; on an IOC you often cannot anyway, and if you could you would be feeding a fault.

### Overvoltage (DC bus high)

**What it means:** the DC bus voltage climbed above the drive's high limit. The bus normally sits at roughly 1.35 times the incoming AC line voltage (so a 480 V line gives a bus around 650 V DC; check your drive's rated bus). Overvoltage means something pushed it higher.

**What causes it:**

- Regeneration on deceleration. This is the classic. When you decelerate a spinning load, the motor becomes a generator and pumps energy back into the DC bus. High-inertia loads — big fans, flywheels, centrifuges — do this hard. If the drive has no way to burn that energy, the bus voltage rises and trips. It almost always trips *on decel*, which the fault snapshot will show you.
- Deceleration too fast. Same mechanism, made worse by a short ramp. Lengthening the decel ramp is a legitimate fix here, because the cause really is that you are asking the load to give up its energy faster than the bus can absorb it.

- An overhauling load. A load that drives the motor — a downhill conveyor, an unwinding spool, a load being lowered — regenerates continuously, not just on decel. That needs a real place to put the energy.
- No dynamic braking where the application needs it. If the load regenerates, the drive needs a dynamic braking resistor (a resistor that burns the returned energy as heat) or a regenerative/line-regen front end. Missing or failed braking resistor, or a failed braking transistor, shows up as overvoltage on decel.
- High incoming line. If the plant line is running hot, the bus starts high and has less headroom. Measure the incoming line voltage against nameplate and against the drive's input rating.

**How to check:** Read the snapshot — decel or overhaul? Measure the incoming line at the drive input with a true-RMS meter; compare all three phases and compare against the drive's rated input window. If a braking resistor is fitted, check it for an open circuit (with the bus verified dead) and check the braking transistor. Lengthen the decel ramp if the application allows it. Do not just widen the trip; the energy has to go somewhere, and "somewhere" is a resistor, not a parameter.

### Undervoltage (DC bus low)

**What it means:** the DC bus dropped below the low limit. The drive trips to avoid running the motor on a starved bus, which would overheat it.

**What causes it:**

- Supply sag. The incoming line dipped — a big load started nearby, a utility event, a transformer that is undersized for the peak. If the drive trips undervoltage at the same moment another large motor across the plant starts, that is your answer.
- A loose lug or connection. A loose incoming power terminal drops voltage under load and heats up. This is common and it is dangerous because loose lugs cook, discolor, and eventually arc. Look for heat discoloration on the input terminals (with power off and verified).
- A blown input fuse or lost phase. Lose one of the three input phases — blown fuse, tripped upstream device, failed contactor pole — and the bus cannot hold up under load. Single-phasing also stresses everything downstream.

**How to check:** With appropriate PPE, measure all three incoming phases at the drive input under load, phase-to-phase. They should be balanced and within the drive's input window. A phase reading low or zero means a lost phase — chase the fuse, the disconnect, and the upstream feeder. If all three are fine at the drive but the bus still sags, look at connections between the metering point and the bus. Torque input lugs to spec after verifying dead; a loose lug is the quiet cause behind a lot of "random" undervoltage trips.

## Ground fault

**What it means:** the drive detected current flowing to ground on its output — the sum of the three output phase currents did not come out to zero, which means current is leaking somewhere it should not.

**What causes it:**

- Motor or cable insulation breakdown. A winding gone to ground, a cable nicked against a gland or tray, a splice let go.
- Moisture, especially in a corrosive or washdown plant. Water in a motor junction box, condensation in conduit, a flooded pull box. In a wastewater or biogas plant, the corrosive atmosphere eats insulation and connections over time, and moisture bridges what is left. This is one of the most common real ground faults I find in those environments.

**How to check — the insulation (megger) test, done right:** This is where people hurt equipment by doing it wrong. LOTO the drive, verify the DC bus is dead, and *disconnect the motor leads from the drive output terminals.* You must isolate the drive before you megger, because a megohmmeter applies hundreds of volts DC and will damage the drive's output electronics if it is still connected. With the drive isolated, megger each motor lead to ground, one at a time, at the voltage appropriate to the motor (commonly 500 V or 1000 V DC for a 480 V motor — follow the motor manufacturer and your site standard). A healthy motor and cable read high — typically many megohms; the old rule of thumb is at least one megohm per kilovolt of rating plus one, but treat that as a floor and trend against the machine's own history. A reading in the low kilohms, or dropping while you watch, is a ground fault. If the motor megs fine at the motor with the cable disconnected but bad with the cable, the cable is your fault. Dry it, find the water path, and fix the insulation — do not just reset and run, because a ground fault in a classified gas area is an ignition source.

## Overtemperature (drive)

**What it means:** the drive's own heatsink or internal temperature sensor exceeded its limit. The power devices are getting too hot. This is the drive protecting itself, and ignoring it kills drives.

**What causes it, in likely order:**

- Blocked cooling. Clogged filter mats, dust-caked heatsink fins, blocked vents. In dusty or fibrous plants this is the number one cause and it is entirely maintenance.
- Failed cooling fan. The drive's internal fan or the enclosure fan quit or is running slow. You can often hear it, or not hear it. Many drives log a separate fan fault.
- Ambient and enclosure heat. Summer afternoon, a sealed enclosure with a dead cabinet fan, a drive mounted next to a heat source, or a room that has no ventilation. This is the one that gives you the "trips every afternoon" pattern.
- Too much load pushing the drive hard continuously, which raises device temperature even when everything else is clean.

- Carrier (switching) frequency set too high for the frame. The carrier frequency is the rate the drive switches its output transistors; raising it quiets the motor whine but adds switching losses and heat in the drive. If someone bumped it up for a noise complaint, the drive can run hot at a load it used to handle. Most manufacturers require you to derate the drive's current rating as carrier frequency climbs — check the derating table for this drive.

**How to check:** Look and feel. Are the filters clean? Is the fan turning? Is the enclosure hot inside relative to the room? Check the drive's ambient rating and derate for altitude if you are up in the Sierra foothills. Check the programmed carrier frequency against the drive's derating table for the running load. Clean or replace filters, confirm the fan, and confirm the enclosure cooling actually works. Do not defeat the thermal trip; it is protecting silicon you cannot replace at 2am.

### Motor overload (thermal)

**What it means:** the drive's electronic thermal overload model decided the motor has been drawing too much current for too long and is overheating. This is a slow, integrating trip — it tolerates a short overload and trips on a sustained one. It is different from IOC, which is instantaneous.

**What causes it:**

- A real mechanical load problem. The machine is genuinely working too hard — a dragging belt, a misaligned coupling, a worn bearing adding drag, a pump running off its curve, product buildup, a fan with a restricted damper. The overload is telling you the truth.
- Wrong motor parameters. The drive's overload model is only as good as the motor FLA and thermal settings you gave it. If someone entered the wrong nameplate FLA, or the drive was swapped and never reprogrammed for this motor, the model can trip early. Verify the programmed FLA against the *actual motor nameplate* — this is a legitimate correction, not the same as widening the limit.
- Belt or coupling issues, low-speed cooling. A motor run at low speed for long periods with a shaft-mounted fan does not cool as well and can overheat at a current that would be fine at full speed; some applications need a separate blower (a constant-speed cooling fan) on the motor.

**How to check:** Compare the drive's running output current to the motor nameplate FLA. If the motor is genuinely pulling near or over FLA at steady state, the load is the problem — go find the mechanical cause, LOTO first, and check alignment, bearings, belt tension, and what the machine is actually moving. If the drive's programmed FLA does not match the nameplate, correct it to match the nameplate. What you do *not* do is bump the overload class or FLA above the motor's real rating to stop the trips. That is how you cook a winding, and on a hot motor in a gas area you have now added an ignition risk. The overload exists so the motor does not become a heater.

### Loss of communications / reference

**What it means:** the drive lost the signal telling it what to do — either the network command (from a programmable logic controller, PLC, over a fieldbus) or the analog speed reference (a 4-20 mA or 0-10 V signal). Most drives fault or go to a programmed fail-safe when the reference disappears.

**What causes it:**

- Network fault. A dropped fieldbus, a failed switch, a bad patch cable, a duplicate address, a PLC that stopped scanning. Look at the network status LEDs on the drive and the error counters if the drive exposes them.
- Analog reference noise or loss. A 4-20 mA loop reading zero (broken wire, dead sensor, loose terminal) or an analog signal getting corrupted by noise. Long analog runs pulled next to VFD output cables or big contactors pick up electrical noise and make the speed reference jump around, which can look like erratic operation or a reference-loss trip.
- Shield and grounding problems. The cure for reference noise is almost always the shield. The signal cable shield must be grounded at *one* end (usually the drive/panel end), not both, or you create a ground loop that injects the very noise you are trying to keep out. A shield grounded at both ends, or not grounded at all, is a classic cause of noisy analog references and intermittent comms.

**How to check:** For a network drop, check the drive's comms status indication, the physical cabling, and whether the PLC is actually commanding. For analog, meter the reference signal at the drive input terminals and watch it — is it steady, or is it dancing? Trace the loop for a broken wire or dead source. Check that the signal cable is shielded, separated from power cabling, and grounded at one end only. Noise problems that come and go with other equipment running are almost always shield and separation problems.

## Symptom → likely cause → check

| Fault class | Symptom / when it trips | Likely cause | Check (and log) |
|---|---|---|---|
| Overcurrent (IOC) | Trips instantly on start, before rotation | Short/ground in cable or motor | Isolate drive, LOTO, verify bus dead, disconnect leads, megger motor and cable |
| Overcurrent (IOC) | Trips at start of accel, load won't move | Mechanical bind or jam | LOTO, turn shaft by hand, inspect load for jam/seizure |
| Overcurrent (IOC) | Trips during accel on a heavy load | Accel ramp too fast (rule out bind first) | Compare accel time to load inertia; lengthen ramp only after mechanical is cleared |

| Fault class | Symptom / when it trips | Likely cause | Check (and log) |
|---|---|---|---|
| Overvoltage | Trips on decel, high-inertia load | Regen with no/failed dynamic braking | Read snapshot; check braking resistor (open?) and transistor; lengthen decel ramp |
| Overvoltage | Trips while lowering/overhauling | Continuous regeneration | Confirm braking scheme suits an overhauling load |
| Overvoltage | Trips randomly, bus starts high | High incoming line | Measure line vs. drive input rating |
| Undervoltage | Trips when a big load starts nearby | Supply sag / undersized feeder | Measure all 3 phases under load; check upstream sizing |
| Undervoltage | Random, terminals discolored/warm | Loose input lug | Verify dead, inspect for heat marks, torque lugs to spec |
| Undervoltage | Trips under load, one phase off | Blown fuse / lost phase | Meter all 3 input phases; find the missing one |
| Ground fault | Trips on start, wet/corrosive plant | Insulation breakdown, moisture | Isolate drive, megger each lead to ground; find water path |
| Overtemp (drive) | Trips on hot afternoons | Ambient/enclosure heat, dead cabinet fan | Check enclosure temp, cabinet fan, ventilation, derating |
| Overtemp (drive) | Trips after running a while, dusty plant | Blocked filter/heatsink or failed drive fan | Inspect and clean filters/fins; confirm fan spins |
| Motor overload | Slow trip after minutes of running | Real mechanical load (drag, alignment, bearings) | Compare running current to nameplate FLA; LOTO and inspect load |
| Motor overload | Trips early, drive recently swapped | Wrong FLA/parameters | Verify programmed FLA against motor nameplate; correct to nameplate |
| Loss of comms/ref | Trips when network drops | Fieldbus/switch/cable/PLC scan | Check comms LEDs, cabling, PLC command |

| Fault class | Symptom / when it trips | Likely cause | Check (and log) |
|---|---|---|---|
| Loss of comms/ref | Erratic speed, trips near other gear running | Analog noise / shield issue | Meter reference at terminals; fix shield ground (one end), separate from power |

## 10-minute triage

Work top to bottom. Stop when the code tells you where to go.

1. **Minute 0-2: Read the fault and the log.** Get the exact code and what this drive's manual calls it. Note the timestamp, whether it was accel / decel / steady, whether it was loaded, and whether there's a pattern (time of day, coincident with other equipment). Write it down before touching anything.
2. **Minute 2-4: Classify.** Is it a *current* fault (overcurrent, overload, ground), a *voltage* fault (over/undervoltage), a *thermal* fault (drive overtemp), or a *signal* fault (comms/reference)? Each sends you a different direction.
3. **Minute 4-6: Look and listen at the keypad, unpowered where needed.** Is the enclosure hot? Fan turning? Filters clogged? Any burnt smell, discoloration, or moisture? Are the comms/network LEDs healthy? These cost nothing and often end the search.
4. **Minute 6-8: One safe measurement that fits the class.** Voltage fault → meter the three incoming phases under load (PPE on). Comms/reference → meter the analog reference or read the network status. Current or ground fault → do NOT meter live output; plan the isolate-and-megger instead.
5. **Minute 8-10: Decide the safe next step, not the quick reset.** If it's current/ground: LOTO, verify the DC bus is dead, disconnect motor leads, megger. If it's overtemp: clean and confirm cooling. If it's overvoltage on decel: check braking and consider a longer decel ramp. If it's overload: compare running current to nameplate FLA and go find the load. If it's comms: fix the cable, the shield, or the source.

Never close out a VFD trip by widening the trip. If you widen an overload or a current limit to make the fault stop, you have removed the protection and left the cause running. The motor, the drive, or the machine pays for that later, and in a gas or dusty plant "later" can mean a fire.

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A note on getting help. On a VFD, the ugliest faults are the ones that get handed between the electrician who owns the panel, the motor shop that owns the winding, and the integrator who owns the network. Each one tests their piece, finds it clean, and hands it back. An independent second look — vendor-neutral, reading the same fault log across all three domains — usually finds the seam they were passing over. That is the fault I get the most calls about.